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RESEARCH ARTICLE

AI-Driven Analysis and Mitigation of Control-Plane Signaling Anomalies in Next-Generation Mobile Networks

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ABSTRACT Control-plane signaling plays a critical role in mobility management, session control, and resource allocation in mobile communication networks. However, the increasing reliance on cloud-native architectures and service-based interfaces in 5G and emerging 6G systems has made the control plane more vulnerable to signaling anomalies, such as signaling storms and Application Program Interface(API) flooding, which can exhaust network resources without corresponding user-plane traffic. This article presents an integrated analytical and AI-assisted framework for detecting and mitigating control-plane signaling anomalies in next-generation mobile networks. Radio Resource Control behavior is modeled using a continuous-time Markov chain that captures state transitions, virtualization overheads, and queueing effects in core network functions. A high-fidelity, event-driven digital twin is developed to emulate realistic interactions between the radio access network and the cloud-native core under both benign and adversarial conditions. Anomaly detection combines analytical likelihood scoring with machine learning models to improve robustness and reduce false alarms. Mitigation is formulated as a constrained decision problem and addressed using safe reinforcement learning to regulate signaling activity while preserving service quality. Simulation results with large-scale user populations demonstrate significant reductions in peak signaling load and core-network queueing delay, highlighting the effectiveness of the proposed framework for enhancing control-plane resilience in future 5G and 6G networks.

INDEX TERMS 5G, 6G, signaling storms, control plane, anomaly detection, digital twin, reinforcement learning.

I. INTRODUCTION

The proliferation of smart mobile devices, IoT sensors, autonomous systems, and data-driven applications has increased the complexity of mobile communication networks and heightened their exposure to cyber threats. A primary challenge for these networks is that greater automation and artificial intelligence, although essential for managing mobility and scaling operations, also increase the risks associated with abnormal or malicious control-plane signaling. Consequently, there is an urgent need to strengthen control-plane security, as automation simultaneously enhances operational efficiency and magnifies security vulnerabilities.

In modern mobile environments, the continuous generation of telemetry and signaling traffic by devices and services significantly affects network reliability. Table 1 lists the abbreviations used in this paper. Signaling surges, whether resulting from poor design, misconfigurations, or compromised devices, can overwhelm control-plane capacity. This situation can lead to service outages or degraded performance for legitimate users and increase the network's susceptibility to covert exploits, thereby underscoring a critical security challenge. The transition from 4G to 5G, along with the development of 6G systems, has substantially expanded the attack surface of the mobile network control plane.

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TABLE 1. List of abbreviations.

Abbreviation	Description
5G / 6G	Fifth / Sixth Generation Mobile Communication System
AI	Artificial Intelligence
CTMC	Continuous-Time Markov Chain
RAN	Radio Access Network
RRC	Radio Resource Control
SBA	Service-Based Architecture
AMF	Access and Mobility Management Function
UE	User Equipment
QoS	Quality of Service

The adoption of cloud-native core architectures, service-based interfaces, network function virtualization, edge computing, and network slicing has created tightly coupled signaling dependencies between the radio access network (RAN) and core network functions. These dependencies increase the vulnerability of modern mobile networks to signaling storms, service-based API flooding, and synchronization-driven control-plane overloads. Figure 1 presents the high-level architecture of the proposed AI-driven control-plane security framework.

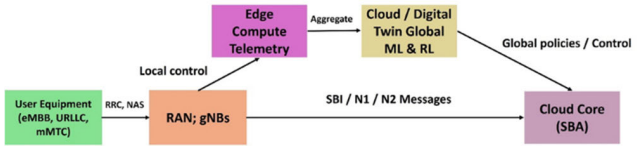


FIGURE 1. High-Level architecture of the proposed ai-driven control-plane security framework.

The proliferation of mobile malware, bot-driven traffic, and automated attack tools, combined with increased user mobility and device density, has resulted in a significant rise in both intentional and unintentional signaling surges. These risks extend beyond service availability degradation to also compromise confidentiality and operational integrity, particularly in critical sectors such as smart cities, autonomous transportation, industrial IoT, emergency services, and digital healthcare. Consequently, protecting the control plane against signaling storms is essential for mobile network operators to ensure infrastructure stability, service continuity, and user satisfaction. As networks evolve toward intelligent 5G and future 6G architectures, which are defined by massive machine-type communications, ultra-reliable low-latency services, and autonomous control loops, manual intervention becomes increasingly impractical. Future mobile networks will depend on self-adaptive, AI-driven mechanisms to monitor, detect, and mitigate signaling anomalies in real time. This shift underscores the necessity for rigorous analytical models and realistic evaluation platforms that accurately represent the interactions among control-plane protocols, virtualization overheads, and AI-based network management. In response to the demand for advanced and adaptive network protection, this paper introduces an integrated analytical and data-driven framework for analyzing, detecting, and mitigating control-plane signaling anomalies in next-generation mobile networks.

The principal contributions of this study are as follows:

A. ANALYTICAL MODELING OF MODERN 5G/6G SIGNALING DYNAMICS

We develop a probabilistic signaling model based on a continuous-time Markov chain (CTMC) formulation to capture RRC state transitions, synchronization-driven signaling bursts, virtualization overheads, and queueing effects across cloud-native core functions such as AMF, SMF, and UPF. The model extends classical 3G/4G signaling analyses to modern service-based architectures.

1) DATA-DRIVEN PARAMETER ESTIMATION FOR REALISTIC SYSTEM BEHAVIOR

A telemetry-driven estimation framework is introduced to derive signaling arrival rates, service rates, transition delays, and synchronization characteristics from observed network traces. These parameters are continuously updated using lightweight online techniques, allowing the analytical model to remain calibrated under dynamic traffic conditions. The average arrival rate is estimated from inter-arrival times as shown in equation 1.

$$\hat{\lambda} = \frac{N}{\sum_{i=1}^{N-1} (t_{i+1} - t_i)} \tag{1}$$

N = number of observed arrivals

t_i = telemetry time of the ith arrival where N denotes the number of observed arrivals and t_i represents the telemetry timestamp of the ith arrival. This formulation supports real-time model calibration using exponentially weighted moving averages under evolving system behavior.

B. HIGH-FIDELITY DIGITAL-TWIN SIMULATION FRAMEWORK

We design a scalable, event-driven digital twin that emulates end-to-end signaling interactions among thousands of user devices, gNBs, and cloud-native network functions. The simulator integrates realistic application traffic, mobility behavior, microservice queueing, and adversarial signaling patterns to study both transient and steady-state anomalies. The evolution of microservice queue states is modeled as shown in equation 2.

$$\frac{dQ_j(t)}{dt} = \sum_{u=1}^{U} \alpha_{u,j} \lambda_u(t)\, p_{u\to j}(t) - \mu_j(t)\, \mathbf{1}_{Q_j(t)>0} + \eta_j^{\text{mob}}(t) + \eta_j^{\text{atk}} \tag{2}$$

where $Q_j(t)$ denotes the queue length of microservice j, U is the number of user devices, $\lambda_u(t)$ is the signaling arrival rate from user u, $\alpha_{u,j}$ is the routing weight, $p_{u\to j}(t)$ captures mobility and connectivity effects, $\mu_j(t)$ is the service rate, and $\eta_j^{\text{mob}}(t)$ and $\eta_j^{\text{atk}}(t)$ represent mobility-induced and attack-induced signaling bursts, respectively.

C. AI-DRIVEN DETECTION USING HYBRID MODEL–ML FUSION

A multi-layer detection pipeline is proposed that combines analytical likelihood scoring derived from the CTMC model with machine-learning techniques, including autoencoders, sequence models, and graph neural networks. A fused scoring mechanism reduces false positives while enabling robust detection across RAN, core-network, and service-based signaling layers.

- ***Safe Reinforcement Learning for Adaptive Mitigation*** Signaling mitigation is formulated as a constrained decision problem and addressed using a safe reinforcement-learning strategy based on actor–critic methods. The proposed policy adaptively adjusts signaling timers, throttles anomalous requests, and triggers controlled scaling actions while respecting quality-of-service and fairness constraints.
- ***Digital Twin Validated Action Safety*** Prior to deployment, each reinforcement-learning action is validated within the digital twin using short-horizon predictive simulations, enabling risk-free testing and preventing unintended quality-of-service violations.
- ***Thorough Experimental Analysis*** Extensive simulations involving 10,000 heterogeneous user devices, 18 gNBs, and a service-based 5G core demonstrate that the proposed framework reduces peak signaling load by up to 45% and lowers AMF queueing delay by approximately 50%, while preserving application-level quality of service.
- ***Anticipatory 6G Signaling Security Insights*** The paper identifies emerging vulnerability patterns related to cross-plane interactions, edge–cloud signaling, and AI-driven control loops, providing guidance for robust control-plane security architectures in future 6G networks.

Overall, this work establishes a foundational framework for understanding and mitigating signaling storms through a combination of analytical modeling, digital-twin simulation, and AI-driven control. The remainder of this paper is organized as follows. Section I-A1 reviews related work on control-plane signaling anomalies and AI-based mitigation techniques. Section II presents the system model and analytical formulation. Section II-A describes the digital-twin simulation environment and traffic and attack models. Section II-B discusses the experimental setup and results. Finally, Section II-C concludes the paper and outlines future research directions.

II. LITERATURE REVIEW

The literature demonstrates the significant impact of Artificial Intelligence (AI) technology on the analysis and mitigation of control-plane signaling anomalies in the evolution of mobile networking technologies. Figure 2 illustrates the connection graph of key research directions in signaling anomaly detection. As mobile communications transition from Fourth Generation (4G) to Fifth Generation (5G) and potentially to Sixth Generation (6G) technologies, the control plane remains one of the most fragmented systems, spanning from the network core to individual slices within service-oriented architectures (SBA) and to disaggregated components in O-RAN and Advanced RAN (Radio Access Network) systems [1], [2]. These systems operate within highly dynamic and heterogeneous network environments, where control-plane signaling must maintain high levels of trust, reliability, and performance.

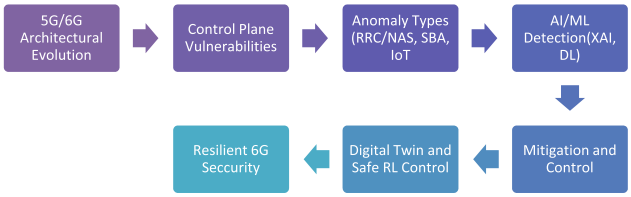


FIGURE 2. Connection graph of key research directions in signalling anomaly detection.

Control-plane signaling anomalies, including signaling storms, protocol failures, abuses, and sophisticated stealth attacks, significantly disrupt network operational availability, functionality, anomaly management integrity, and overall quality of service (QoS) [3], [4]. Traditional detection approaches, such as rule-based and signature-based methods, have proven inadequate for these evolving systems. The increasing complexity and diversity of protocol-based signaling storms, multi-vendor deployments, and adaptive protocol behaviors challenge the scalability of conventional detection systems [5], [6]. Consequently, Artificial Intelligence (AI) and Machine Learning (ML) have emerged as viable alternatives for control systems. Advanced ML techniques can adaptively learn and identify anomalies in modern networks, offering a more effective solution than static, rule-based approaches [7], [8].

This section examines publications from 2020 to 2025, with a focus on advanced techniques for detecting and managing control-plane anomalies in 5G, O-RAN, and early 6G systems, utilizing recent advancements in AI technology. The review covers a spectrum of methods, including basic statistical approaches, deep learning techniques, federated learning systems, and explainable AI, and identifies persistent gaps in the operational implementation of these approaches.

A. TECHNIQUES FOR ANOMALY DETECTION IN THE CONTROL PLANE USING AI/ML

Yao and Chen [1] recently investigated anomaly detection in Open Radio Access Networks (O-RAN), employing

Explainable Artificial Intelligence integrated with statistical models to process O-RAN telemetry data such as RRC connections, users' geographical histories, and handover data. By utilizing real datasets instead of synthetic traces, the study demonstrates the practical reliability of these detection frameworks in identifying anomalous patterns. Although the data were collected in the field, the authors acknowledge challenges related to operational generalization, particularly due to the heterogeneous and dynamic nature of O-RAN deployments under varying network conditions. Rouili et al. [7] examined 5G O-RAN environments and Intrusion Detection Systems (IDS) using an In-Switch Random Forest (RF) Classifier. Their approach focused on RAs/RRC request predictors and leveraged programmable switches to achieve ultra-low-latency detection. The system attained 94% accuracy with an inference time of 3.4 μs, indicating suitability for real-time operationalization. However, reliance on P4 programmable hardware limits flexibility, as the system can detect only specific attack classes, thereby restricting generalization.

Scalingi et al. [21] introduced an Edge AI framework for threat detection in 5G O-RAN systems, integrating partially available PHY-layer fingerprints, time-of-arrival (TOA) tags, and RRC-level metrics within a PHY-plus-RAN emulator. The framework is designed for near real-time operation, achieving 85% accuracy with 2 ms latency. However, the authors identified a significant gap between emulator-based evaluations and deployment in live networks, which may hinder practical implementation. Oskouei et al. [18] proposed a hybrid anomaly-detection system for mobile networks that combines Deep Belief Networks (DBNs) and advanced transformers to analyze time-series telemetry. Their use of self-attention and temporal feature extraction demonstrates state-of-the-art anomaly-detection capabilities in adaptive models. Nevertheless, the computational expense of this approach limits its feasibility for real-time applications in mobile networks.

Pan et al. [15] introduced FedGAD, a federated GAN-based anomaly detection framework designed for emerging 6G edge environments. The system utilizes a distributed edge dataset and incorporates an adversarial generator alongside multiple local discriminators to enhance detection and adaptability across heterogeneous nodes. This approach demonstrated notable improvements over centralized and non-adversarial benchmarks. However, the work represents an initial proof-of-concept, and its performance in real-world edge scenarios remains largely unverified. Kumar et al. [16] investigated anomaly detection dynamics in 5G slice environments using a ConvLSTM model trained on transformed slice-traffic data. Their method effectively captured spatio-temporal slice features, achieving 99% multiclass accuracy. Nonetheless, concerns regarding dataset replicability may impede subsequent validation efforts.

Ilias et al. [17] implemented a convolutional neural network (CNN) with a Mixture-of-Experts architecture for analyzing 2D traffic matrices in a 5G NIDD testbed, achieving a weighted F1 score of approximately 99.95%, which demonstrates high classification accuracy. Nevertheless, the approach is constrained to a specific testbed, limiting its generalizability to diverse 5G environments. Ruba et al. [11] developed a Federated LSTM Autoencoder for anomaly detection in distributed cellular access networks. Leveraging time-series infrastructure telemetry, the framework delivered competitive detection performance while reducing centralized training time. However, the study identifies communication-overhead trade-offs as a persistent challenge in federated learning architectures. Pirbhulal et al. [25] proposed a hybrid graph neural network (GNN)–CNN architecture for real-time anomaly detection in 5G-IoT networks. By utilizing graph-structured features and flow-level statistics from live 5G data, the model achieved 99.19% accuracy for binary classification. However, the focus on binary attack detection limits its applicability to multi-class or complex threat scenarios.

Dayaratne et al. [8] proposed the use of large language model (LLM)-based xApps to enhance durability within O-RAN service development layers (SDL). The model, evaluated with modified hypoglyph inputs, demonstrated very low latency (less than 0.07 seconds) and notable robustness. However, further improvements are required to achieve sufficient accuracy for deployment in real-world O-RAN environments. Gurushanker et al. [4] evaluated multiple machine learning classifiers for anomaly detection in the 5G core, utilizing TCP/IP and PFCP traffic captures. The classifiers achieved high accuracy (approximately 97%) for TCP/IP flows, but accuracy decreased to 69% for PFCP

TABLE 2. Technique VS capability matrix.

Ref.	Study	Federated Learning	Cross Layer Detection
[1]	Partial (near-RT)	✓	High (operational O-RAN)
[2]	Partial (testbed)	✓	High (real testbed)
[3]	✓	✗	Medium (telemetry)
[4]	Partial	✗	Medium (core testbed)
[6]	✓ (simulation)	✗	Low (simulation)
[7]	✓	✗	Medium (O-RAN testbed)
[8]	✓	✓ (SDL-layer)	Medium (manipulated SDL)
[9]	✓	✗	Medium
[10]	✓	✗	Very high (global IoT)
[11]	Partial	✗	Medium (distributed telemetry)
[12]	Partial	✓	Medium (FL datasets)
[21]	✓	✓ (PHY + RRC)	Medium (emulator)
[14]	✓	✗	Medium (slice data)
[15]	✓	✗	Medium (edge datasets)
[16]	Partial	✗	Low–medium (slice images)

data streams. This outcome highlights the distinct challenges associated with control-plane protocols and the variability in protocol-dependent performance. Dai et al. [6] developed an anomaly detection model based on state changes and sequence behaviors using 5G control-plane data simulated with a Markov Chain. Although the method performed well in controlled simulations, the absence of real-world validation and reliance on strong model assumptions undermine confidence in its operational viability. Ganiuly and Bolatbek [14] introduced per-slice machine learning models for anomaly detection in 5G slices, utilizing features such as slice IDs, QoS class, and resource usage. The models achieved strong performance, with M1 class scores ranging from 0.93 to 0.96. However, the inability to maintain separate models for each slice increases the complexity of anomaly assessment per slice.

Omaña Iglesias et al. [10] introduced an unsupervised hybrid statistical and deep learning approach for anomaly detection on an international IoT roaming platform. By leveraging connectivity patterns and device behavioral signatures, the method demonstrated improvements in operational filtering. However, the absence of comprehensive performance metrics limits the transparency and replicability of the results. Wen et al. [19] employed cross-layer features for edge-level O-RAN attack trace anomaly detection, integrating explainable artificial intelligence (XAI) and large language models (LLMs). This approach achieved up to 100% detection for a subset of attacks. Nevertheless, the limited diversity of attacks in the dataset raises concerns regarding the generalizability of the findings to broader O-RAN scenarios.

Sun et al. [20] introduced an explainable artificial intelligence (XAI)-based classification framework for security analysis of sixth-generation (6G) non-terrestrial networks (NTNs), with particular emphasis on drone communication systems. Simulations of drone networks demonstrated strong classification performance. Bhalerao et al. [23] developed a model that integrates graph neural networks (GNNs) and reinforcement learning to mitigate risks in fifth-generation (5G) networks, utilizing a graph-based representation of network topology and reinforcement learning for real-time response. This approach represents an initial advancement in AI-based risk assessment. Table 2 presents the Technique versus Capability Matrix. A significant challenge remains in integrating such models with operational and real-world infrastructure. Siniarski et al. [26] outlined a conceptual roadmap for trustworthy and explainable AI in 6G security, emphasizing multidimensional trust, transparency, and resilience. As a vision paper, it currently offers limited empirical analysis and lacks real-world system applications.

B. COMPARISON OF ANOMALY TYPES AND DETECTION APPROACHES

Gurushanker et al. [4] evaluated machine learning-based classification of PFCP signaling-plane attacks across both PFCP and TCP/IP flows. The models achieved 97% accuracy on TCP/IP datasets, demonstrating strong performance. In contrast, PFCP flows yielded only 69% accuracy, highlighting protocol-specific limitations and challenges in scaling detection methods across diverse control-plane protocols.

Regarding O-RAN RA-based signaling storms, Rouili et al. implemented an in-switch Random Forest Classifier, demonstrating strong scalability and line-rate performance [7]. Scaling et al. [21] proposed an Edge AI solution that achieved 94% accuracy and an inference time of 3.4 μs per flow. Notably, this was the first instance of attack detection using cross-layer RRC/PHY features, achieving 85% accuracy and real-time results with 2-millisecond latency. While these results suggest suitability for Edge Analytics, the evaluation was limited to emulator-based environments, which constrains conclusions about live deployment.

Dai et al. [6] utilized Markov models to simulate control-plane anomalies, but the model has not been validated due to the need for accurate estimation of real network conditions. Although real-time calculations are feasible, the scalability of predictions depends on specific scenarios. Ganiuly and Bolatbek [14] introduced a modified slice machine learning model, achieving F1 scores of 0.93, 0.94, and 0.96. This model is partitioned for slice-level scalability and real-time performance on edge hardware, requiring deployment of multiple models for different slices. For IoT device anomaly detection, Omaña Iglesias et al. [10] proposed an unsupervised hybrid statistical deep learning methodology. While this approach enables filtering of IoT devices on a global roaming platform, it demonstrates minimal performance metrics, particularly in terms of production latency required for operational deployment.

Wen et al. [19] integrated fully and partially explainable anomaly detectors with large language models (LLMs) to identify edge-level attacks within O-RAN. Their prototype provided runtime explanations and achieved 100% positive detection for certain attacks; however, its scalability to broader attack types beyond small, curated datasets remains unproven. Parada et al. [2] developed deployable experimental O-RAN anomaly-detection applications, evaluating them on authentic testbeds. This work demonstrates potential for systematic scalability and deeper integration with the O-RAN architectural framework, leveraging repeatable, time-based metrics from testbed experiments.

Tymokhin et al. [9] AI fraud detection on Interconnect bypass by analyzing call control behavioral patterns. Their approach was low-code and real-time and has the potential to be useful throughout the entire interconnect telecoms ecosystem. Pirbhulal et al. [25] designed an anomaly detection system for 5G-IoT and achieved real-time operation with 99.19% binary classification accuracy, using a hybrid GNN–CNN. The system's environment was within the limits of IoT scale, and it supported additional optimizations that enabled deployment in larger-scale settings; however, it was only capable of binary classification. Dayaratne et al. [8] suggested using xApps based on LLMs to uncover O-RAN-manipulated data in a hypoglyphic style without bypass.

Although the system is highly performant in terms of latency (<0.07s), it still does not meet the accuracy required for deployment in xApps at scale. Kumar et al. [16] trained a ConvLSTM model for detecting slice-traffic anomalies, achieving 99% accuracy across all classes. Although the results suggest the possibility of slice-level scalability, the factors of latency and deployment are left unexplored. Ilias et al. [17] reported a weighted F1 score of 99.95% using a CNN with a Mixture-of-Experts architecture for intrusion detection in a 5G-NIDD network. The results are promising, yet validation was performed primarily on a controlled testbed, which limits the extent of claims about scalability for the research.

C. MITIGATION STRATEGIES AND THEIR EFFECTIVENESS

Rouili et al. [7] implemented a machine learning-based anomaly detection method for programmable switches, utilizing Tofino ML-based programmable switch hardware. The resulting system demonstrated real-time performance, achieving a 3.4 μs inference time and 94% detection accuracy, and is considered scalable with appropriate hardware. Scalting et al. [21] further examined scalability by introducing an Edge-quarantine approach, which achieved 2 ms latency, robust emulation, and real-time response at the Edge/Near-RT RIC. This approach is indicative of effective on-Edge anomaly detection and mitigation, although only moderate scalability was achieved due to the complexity of edge integration.

Chandran and Venkataraman [3] introduced LHB-X, a lightweight heartbeat mechanism at the NWDAF/telemetry level featuring O(1) update capability. This mechanism enhances selective filtering and reduces monitoring overhead; however, its scalability is primarily determined by protocol-level adoption and deployment constraints. Table 3 presents a performance comparison of relevant literature. Allaw et al [22] proposed a hybrid SDN/NFV and AI orchestration framework for control-plane security, which operates near real-time and demonstrates significant improvements in theoretical performance assessments. The scalability of this framework depends on the integration effort required to unify diverse virtualization and orchestration systems. Mavale et al. [24] developed a deep learning-based IDS/IPS system integrated with SDN, aimed at enabling rapid enforcement of network layer policies.

The system is reported to achieve high detection accuracy with statistically negligible false-positive rates. However, its scalability depends on the complexity of runtime SDN policy management. Pan et al. [15] described advances in federated adversarial training across distributed nodes. Their approach incorporates edge-level anomaly detection and model updates. This method enhances defenses against diverse attacks and patterns, but requires careful orchestration of federated learning cycles for scalable deployment. Nugraha et al. [13] emphasized the use of dataset-driven tuning to improve the realism of evaluation and analysis

TABLE 3. Performance comparison of literature.

Ref.	F1 Score	Model Type	Dataset Type
[1]	Field-validated, high	XAI + Statistical Learning	Real O-RAN telemetry
[2]	Testbed-validated	AI-enabled O-RAN apps	Real O-RAN testbed
[3]	Not reported	Lightweight Boolean Signaling	Telemetry
[4]	TCP/IP = 97%, PFCP = 69%	ML classifiers	Core testbed
[5]	High accuracy (Fed LLM)	Federated LLM	Federated 5G data
[6]	Simulation-validated	Markov Chain	Simulation
[7]	≈94%	In-switch Random Forest	P4 switch traces
[8]	High robustness to manipulation	LLM-based	Manipulated SDL
[10]	Operational filtering	AI on Call Control	Telecom CDRs
[11]	Competitive	Statistical + DL	Global roaming IoT
[12]	High FL accuracy	Federated LSTM Autoencoder	Distributed telemetry
[21]	Multiple baselines	Distributed ML + FL	Federated sets
[14]	F1 = 0.93–0.96	ML/DL baseline	Large real 5G dataset
[15]	Operational filtering	Lightweight ML	Slice-level data

in research and testbeds. Wen et al. [19] introduced the edge-analysis O-RAN XAI pipeline, the first system to integrate explainable artificial intelligence (XAI) with anomaly detectors and large language models (LLMs). This pipeline detects 100% of specified attacks and provides instance-level explanations. However, its scalability is constrained by the computational demands of generating explanations. Bhalerao et al. [23] proposed adaptive 5G policy orchestration using graph neural networks (GNN) and reinforcement learning (RL) for risk management. While this enables dynamic risk mitigation, it is limited by the complexity of large-scale RL training, policy integration, and layered mitigation.

Nomikos et al. developed a Distributed Trustable Engine that integrates privacy-preserving anomaly detection with federated learning; its scalability is moderate due to the overhead from distributed coordination and the synchronization required for federated update cycles. Similarly, Tymokhin et al. designed a real-time AI-based fraud detection system for call-control layers, leveraging a low-code architecture that simplifies deployment and enables effective anomaly detection, resulting in a scalable system for telecommunications environments.

D. DATASET CHARACTERISTICS AND EXPERIMENTAL SETUPS

Nugraha et al. [13] were the first to document a control-plane 5G dataset collected from a functioning 5G testbed. This dataset is comprehensively documented and supports both validation and benchmarking. Gurushanker et al. [4] accessed TCP/IP and PFCP captures from a controlled 5G

core testbed, providing a dataset that included both attack and benign flows for evaluation across multiple protocol sets. Ilias et al. [17] utilized attack and benign flows from a real 5G-NIDD testbed, made their dataset publicly available, and evaluated it using a standard train-test split, reporting the weighted F1 score on the test set.

Pirbhulal et al. [25] differentiated benign and malicious flows from real-time 5G traffic collected from a live 5G-IoT network, providing a dataset suitable for binary classification metrics in operational environments. Ruba et al. [11] employed distributed infrastructure telemetry from cellular access networks, consisting of extensive time-series data, to detect anomalies using both unsupervised and federated learning approaches.

Scalingi et al. [21] utilized large-channel and Open RAN emulators to generate emulated data with targeted attack injections, presenting results from both the emulator and prototype validation. Kumar et al. [16] transformed traffic from a single 5G slice into images to detect multiclass slice anomalies, employing train-test splits across various sliced-network scenarios to assess multiclass prediction metrics. Wen et al. [19] leveraged the O-RAN edge testbed to obtain targeted edge-attack traces for model evaluation; however, the small and highly curated dataset supported end-to-end prototype validation rather than large-scale benchmarking. Dai et al. [6] generated and analyzed synthetic traffic in a simulated 5G control-plane environment by sequentially inserting specific anomalies, with validation conducted entirely in simulation and focused on modeled scenarios. Omaña Iglesias et al. [10] employed real multi-operator data from a global IoT roaming platform, featuring large-scale device connectivity and active device roaming anomalies, to support deployment validation.

Yao and Chen [5] analyzed operational O-RAN telemetry data from a production network, which included real-world operational anomalies and enabled field-level validation. Rouili et al. [7] conducted detection validation on purpose-built hardware by recreating storm scenarios on 5G O-RAN testbeds using high-throughput programmable switches.

E. EMERGING TRENDS AND FUTURE DIRECTIONS

Wen et al. [19] introduced explainable edge security for O-RAN by employing large language models (LLMs) to provide real-time explainability of edge-level attacks. Their findings support the need for expanding coverage to additional attack types and further integrating LLMs to enable scalable, explainable security at the network edge. Siniarski et al. [26] proposed a framework for trustworthy AI in 6G systems, emphasizing tailored explainability, robust models, and energy-efficient 'green AI.' Their perspective identifies standardization, resilience, and energy efficiency as essential for reliable and sustainable 6G networks.

Pan et al. [15] implemented Federated GAN–based adaptive anomaly detection to facilitate privacy-preserving training in a distributed 6G edge-computing environment. Their work alludes to a significant 6G rollout in which federated adversarial training will minimize privacy trade-offs for a more robust system. Dayaratne et al. [8] built and designed O-RAN Resilient xApps using LLMs to identify attack scenarios where the system has been provided with duplicitous or masked information. Future work will focus on more nuanced xApp systems able to provide more reliable performance in adversarial scenarios.

Nomikos et al. [12] examined distributed machine learning and federated learning approaches for scalable anomaly detection in 6G networks, analyzing the interplay between privacy-preserving frameworks and cross-network learning. Similarly, Bhalerao et al. [23] presented adaptive policy frameworks utilizing reinforcement learning and graph neural networks (GNNs) for 5G and 6G networks, recommending increased orchestration and automation to support risk-based mitigation strategies.

ChandChandran and Venkataraman [3] introduced LHB-X, a low-bandwidth health indicator function designed to optimize SDT and NWDAF for low-signaling telemetry. Operational data from NWDAF highlights the need for adaptable, scalable health-monitoring systems. Parada et al. [2] developed an open O-RAN testbed that supports AI-based anomaly detection, thereby improving reproducibility and facilitating community-wide comparison of O-RAN tools and datasets. Lingi et al. explores the use of Edge AI and PHY fingerprinting for cross-layer real-time detection ([21]. Their data-driven approach drives the point that shifting from emulators to real networks is necessary for full validation. Sun et al. [20] consider explainable AI in the context of detecting 6G non-terrestrial networks, drones, and protocol violations. The contribution highlights the integration with terrestrial 6G as an essential step toward unified cross-domain anomaly detection in the future.

Allaw et al. [22] proposed a cross-layer slice security architecture leveraging software-defined networking (SDN), network function virtualization (NFV), and AI to enable end-to-end security orchestration. This approach advocates for automated slice-level defenses and enhanced collaboration across network layers. Omaña Iglesias et al. [10] introduced deep learning and unsupervised techniques to establish a scalable framework for anomaly detection in globally connected IoT systems. Future research will address multi-operator collaboration and standardization to support large-scale, interoperable IoT security.

F. CURRENT LIMITATIONS

i) LACK OF DETAILED AND REALISTIC DATASETS

The literature consistently identifies a significant shortage of publicly available, detailed, and realistic datasets for control-plane anomaly detection. Several studies highlight the absence of data for Radio Resource Control (RRC), Non-Access Stratum (NAS), Packet Forwarding Control Protocol (PFCP), and Random Access (RA) control-plane traces [4], [13]. As a result, many investigations depend on simulated data, small-scale testbeds, or proprietary datasets that are not disclosed due to privacy and competitive restric-

tions. This scarcity of scenario-specific data, particularly for RRC and RA [4], [13], impedes the ability of systems to accurately replicate real-world operational network behavior [21], [22], thereby presenting substantial challenges for real-time anomaly detection and documentation. Additional obstacles include the limited diversity of testbeds, which restricts the study of Distributed Denial of Service (DDoS) attacks and other complex events in lower-bandwidth environments [8], [13]. Notably, Nugraha has introduced a more comprehensive 5G dataset that offers broader coverage of both control and data planes [13]. Nevertheless, there remains a lack of community engagement to expand upon these findings and to address a wider range of use cases.

ii) *Complexity of Operational Integration*

Several practical and architectural considerations arise when deploying AI and machine learning-based anomaly detection in operational networks [7]. Key challenges include issues related to in-switch machine learning, particularly the use of programmable switches such as Tofino, which are currently absent in most POD environments. Deep integration is required, as the machine learning approaches proposed by Rouili et al. [7] must also interface with the RAN Intelligent Controller (RIC), Network Data Analytics Function (NWDAF), software-defined networking (SDN) controllers, and network function virtualization (NFV) orchestrators. Interfaces and application programming interfaces (APIs) necessary for automated control remain underdeveloped. Multivendor Open Radio Access Network (O-RAN) environments further complicate integration due to heterogeneous telemetry, machine learning models, and mitigation tools. These integration challenges are exacerbated by limited operational deployment, as most existing literature relies on closed testbeds, with few studies demonstrating results in real-world environments.

Countering telemetry manipulation remains a significant concern. As AI-based detection becomes more prevalent, adversaries increasingly employ tactics such as data poisoning, which alters training data to manipulate models, and adversarial examples, which appear benign to traditional systems but compromise machine learning models [8], [15]. Dayaratne et al. [8] identify hypoglyphic attacks in the O-RAN Service Management and Orchestration (SMO) Data Layer (SDL), where minor data perturbations evade detection, emphasizing the necessity for attack-driven countermeasures. Initial techniques incorporating large language models (LLMs) and adversarial training [8], [15], [19] demonstrate potential but lack comprehensive benchmarking, particularly in operational deployments. Trade-offs between latency and accuracy continue to influence deployment strategies, underscoring the importance of developing realistic AI-based security systems.

Low-latency anomaly detection, with latencies of 1 to a few milliseconds, comes with trade-offs, particularly model simplifications that affect overall accuracy and the detectable magnitude of anomalies. For instance, in-switch deployment relies on simple models, such as compact random forests, that fit into match-action tables due to hardware compilation [7]. These latency-driven limits also affect the features and data processed on the fly, leading to the loss of deep features and a system that is not fast enough for real-time attacks. Thus, navigating the unavoidable trade-off between accuracy and latency is necessary. Every deployment must contend with the fact that ultra-fast anomaly detection excels at basic, non-robust detection, while complex, accurate models take longer. The right balance of these trade-offs depends on the operational context and the network domain's performance needs [7], [21].

Current evidence indicates that AI-based anomaly detection in control planes is transitioning from limited application to broader adoption for securing mobile networks. The ongoing integration of in-network processing, edge AI, federated learning, and explainable AI reflects the development of an ecosystem capable of real-time threat detection and addressing privacy concerns across engineering domains. However, several challenges must be addressed before widespread adoption can occur, including dataset standardization, system integration, robust adversarial design, and energy consumption monitoring. With the advent of 6G, the ecosystem is anticipated to provide highly effective, ultra-low latency, cross-layered defense systems with advanced analytics. Flexible architectures will be necessary to accommodate this complexity. Continued industry collaboration will be essential for developing highly integrated, autonomous, and self-organizing networks that leverage machine learning.

III. SYSTEM MODEL

This section presents the overall system model, including analytical RRC-level modeling, data-driven parameter estimation, a high-fidelity digital twin simulator, and AI-based detection and mitigation mechanisms. The proposed framework records device–user interactions, RRC state transitions, virtualization overheads, and control-plane behavior across cloud-native core components and aligns with state-of-the-art 5G and emerging 6G architectures.

A. NETWORK ARCHITECTURE OVERVIEW

The analyzed 5G/6G mobile network architecture consists of four logical layers. The Radio Access Network (RAN) includes gNBs that interact with User Equipment (UE) through NR interfaces. The RAN implements the RRC state machine and supports the Idle (I), Paging (P), Forward Access Channel (FACH), and Dedicated Channel (DCH) states.

The core network is represented as a cloud-native environment comprising virtualized control-plane and user-plane network functions, such as the Access and Mobility Management Function (AMF), Session Management Function (SMF), and User Plane Function (UPF). These functions operate with finite processing and signaling capacities and are deployed as containerized microservices. The edge layer is responsible for low-latency tasks, including telemetry preprocessing, local anomaly detection, and reinforcement learning

(RL) inference. In contrast, the cloud layer executes computationally intensive machine learning models and hosts the global digital twin for large-scale simulation and policy analysis. Each UE produces both benign and potentially dysregulated signaling traffic. Signaling storms arise when many UEs simultaneously trigger correlated RRC promotions and control-plane procedures, which rapidly deplete radio and core network resources.

B. ANALYTICAL RRC-LEVEL BEHAVIOR MODEL

Each UE is modeled as a continuous-time Markov chain (CTMC) with state space is shown in equation 3.

$$\mathcal{S} = I, P, F_0, F_L, D_0, D_L, D_H \tag{3}$$

where F_0 and D_0 represent inactive substates of FACH and DCH, respectively, while F_L, D_L, and D_H correspond to connected substates with increasing bandwidth demand.

Let λ_L and λ_H denote the arrival rates of low- and high-rate sessions, with corresponding service rates μ_L and μ_H. To capture attacker synchronization with RRC demotion events, synchronization parameters τ_L and τ_H are introduced, yielding effective promotion intensities in equation 4.

$$\Gamma_L = \lambda_L + \tau_L^{-1}, \Gamma_H = \lambda_H + \tau_H^{-1} \tag{4}$$

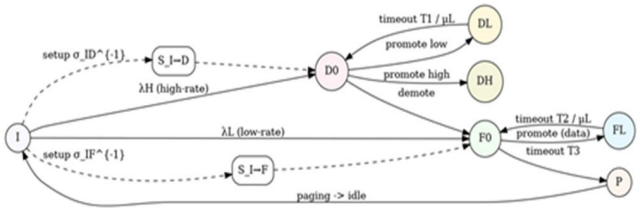


FIGURE 3. RRC CTMC.

Transition overheads associated with resource setup and teardown for a transition $x \rightarrow y$ are denoted by σ_{xy}^{-1}, and the corresponding transient waiting state is denoted S_{xy}. The CTMC balance equations for each state $x \in S$ are given in equation 5.

$$\pi_x r_x^{\text{out}} = \sum_{y \in \mathcal{S} \cup T} \pi_y r_{y \rightarrow x} \tag{5}$$

subject to the normalization constraint shown in equation 6.

$$\sum_{x \in \mathcal{S}} \pi_x + \sum_{x,y} \pi_{S_{xy}} = 1 \tag{6}$$

Solving these equations yields steady-state probabilities π_x, which quantify per-UE state occupancy and form the basis for signaling-load estimation.

C. DATA-DRIVEN PARAMETER ESTIMATION

To enhance model fidelity, CTMC parameters $(\lambda, \mu, \Sigma, \tau)$ are estimated from recorded network telemetry. For each UE u, a timestamped dataset shown in equation. 7.

$$\mathcal{D}_u = \{(t_i, s_i, m_i)\}_{i=1}^{N} \tag{7}$$

is collected, where t_i denotes the event time, s_i the RRC state, and m_i auxiliary metadata (e.g., payload size, slice ID, cell ID). Arrival and service rates are estimated via maximum likelihood as in equation 8.

$$\hat{\lambda}_x = \frac{n_x}{T}, \hat{\mu}_x = \frac{n_x^{\text{end}}}{\sum_i t_x^{(i)}} \tag{8}$$

where n_x is the number of arrivals to state x over observation window T. Transition delays yield empirical estimates in equation 9.

$$\hat{\sigma}_{xy}^{-1} = \frac{1}{|\ T_{xy}|} \sum_i T_{xy}^{(i)} \tag{9}$$

Synchronization parameters are estimated from burst-transition timing offsets as in equation 10.

$$\hat{\tau}_x = \frac{1}{N} \sum_i |\ t_i^{\text{burst}} - t_i^{\text{trans}}\ | \tag{10}$$

Parameters are maintained online using exponentially weighted moving averages (EWMA) is calculated in equation. 11.

$$\theta_{t+1} = (1 - \alpha)\, \theta_t + \alpha \hat{\theta}_t \tag{11}$$

ensuring continuous calibration under evolving traffic conditions.

D. DIGITAL-TWIN SIMULATION FRAMEWORK

A controllable digital twin models both control-plane and user-plane dynamics, enabling realistic simulation of signaling storms and mitigation strategies. User agents follow mobility models and CTMC-based RRC behavior, while AMF, SMF, and UPF are represented as finite-capacity queuing servers with capacities $C_{\text{AMF}}, C_{\text{SMF}}, C_{\text{UPF}}$.

Attack injectors schedule synchronized bursts, botnet-style correlated events, or misbehaving application patterns. The event-driven simulator processes application events, RRC transitions, CNF service completions, and attack triggers. Telemetry includes control-plane load $L_c\,(t)$, queue lengths, transition counts, latency, and QoS metrics. Calibration is validated using distributional tests such as Kolmogorov–Smirnov tests.

E. MACHINE LEARNING BASED ANOMALY DETECTION AND PREDICTION

Anomaly detection follows a layered design combining fast edge-level detectors with cloud-level deep models. Feature vectors include RRC promotion rates, signaling counts, state dwell times, cross-slice correlations, and core-network indicators such as $L_c\,(t)$ and AMF queue length.

Edge detectors employ EWMA and CUSUM for rapid alarms, while cloud models include autoencoders, Deep SVDD, sequence models (LSTM/Transformer), and graph neural networks (GNNs). A CTMC-based likelihood score as shown in equation (12).

$$A_u = -\sum_i \log p\,(s_{i+1} \mid s_i; \Theta) \tag{12}$$

is fused with ML outputs via equation 13.

$$S_u(t) = \alpha A_u(t) + (1-\alpha)\,\mathrm{MLscore}_u(t) \tag{13}$$

reducing false positives while maintaining detection sensitivity.

F. REINFORCEMENT-LEARNING BASED ADAPTIVE MITIGATION

Mitigation is formulated as a constrained Markov decision process. Observations include $L_c(t)$, CNF queue lengths, suspected anomalous UE fractions, and per-slice KPIs. Actions include RRC timer tuning, throttling, temporary quarantine, and CNF scaling. The reward function is shown in equation (14).

$$r_t = -\left(w_1 L_c(t) + w_2 \mathrm{QoS}_{\mathrm{viol}}(t) + w_3 \mathrm{Cost}(a_t)\right) \tag{14}$$

Safe RL policies (e.g., PPO or actor–critic with Lagrangian constraints) enforce fairness and QoS bounds. Before deployment, candidate actions are validated in the digital twin via short-horizon predictive simulations to ensure safe outcomes.

G. REINFORCEMENT-LEARNING BASED ADAPTIVE MITIGATION

In 5G/6G systems, RRC transition latency depends on radio procedures and cloud-native core processing. The total transition delay is modeled as shown in equation (15).

$$T_{xy} = T_{xy}^{\mathrm{radio}} + T_{xy}^{\mathrm{compute}} + T_{xy}^{\mathrm{queue}} \tag{15}$$

The compute delay aggregates processing times across microservices with equation (16).

$$T_{xy}^{\mathrm{compute}} = \sum_{i=1}^{m} \delta_{xy}^{(i)} \tag{16}$$

Queueing delay under an M/M/K approximation is shown in equation (17).

$$T_{xy}^{\mathrm{queue}} = \frac{P_{\mathrm{wait}}}{K\nu - \Lambda} \tag{17}$$

where Λ is the aggregate signaling arrival rate as shown in equation (18).

$$\Lambda = \sum_{u=1}^{M_N} \lambda_u^{(N)} + \sum_{u=1}^{M_A} \lambda_u^{(A)} \tag{18}$$

As $\rho = \Lambda/(K\nu) \to 1$, queueing delays grow sharply, capturing saturation effects during signaling storms. Coupling this delay model with the CTMC steady-state solution characterizes how RRC congestion, CNF overload, and microservice contention jointly determine storm severity in modern 5G/6G networks.

Signaling Transition Latency and Network Load Model for 5G/6G Systems In modern 5G and emerging 6G networks, the signaling delays associated with RRC promotions and demotions are influenced not only by radio access procedures but also by the behavior of cloud-native core functions operating within a service-based architecture (SBA). Unlike legacy

Algorithm 1 Safe Reinforcement Learning–Based Control-Plane Mitigation

Input: UE population, CNF configurations, RRC timers, attack plan, simulation parameters
Output: Recorded traces and metrics ($L_c(t)$, CNF queue lengths, UE RRC timelines, detection/mitigation events, QoS metrics)
Step 1 Initialization
1.1 Set simulation clock $t \leftarrow 0$; create empty event queue Q and log store.
1.2 For each UE u:
•Schedule first application event (sample from app model).
•Set RRC state $s_u \leftarrow$ Idle.
1.3 Instantiate CNF clusters with K replicas and empty queues.
1.4 Initialize attack schedulers (attack_trigger events).
1.5 Set detection and logging timers:
next_detect $\leftarrow d_i nterval$,
next_log $\leftarrow$ l$_i nterval$.
Step 2 Main Event Loop
While $t < T_{\mathrm{sim}}$ **and** Q not empty:
2.1 Pop earliest event e; update $t \leftarrow e.time$.
Event types:
application_event:
•Generate traffic; check if RRC transition needed.
•If so, compute transition latency T_{xy} and enqueue related CNF signaling events.
•Schedule next application event.
cnf_service events:
•On start: dequeue message, schedule completion.
•On completion: update queue stats and forward to next CNF or UE; enqueue follow-up events if needed.
rrc_transition_complete:
•Update UE RRC state and counters.
attack_trigger:
•Inject spurious signaling (fake promotions/bursts).
•Enqueue resulting CNF messages.
Periodic metrics:
2.2 Compute control-plane load $L_c(t)$ and CNF utilization.
2.3 If $t \geq$ next_log: log snapshot; update next_log.
Detection:
2.4 If $t \geq$ next_detect:
•Extract features (promotion rates, signaling counts, queue lengths).
•Call **DetectionAndMitigation()**; schedule mitigation actions if returned.
•Update next_detect.
Step 3 Post-processing
3.1 Save traces ($L_c(t)$, queues, UE states, detection/mitigation events).
3.2 Compute evaluation metrics (ROC, AUC, time-to-detect, QoS impact).
3.3 Return logs and summarized metrics.

Output

Implementing Quality of Service (QoS), fairness, and safe control plane operation trained mitigated RL policy safe-RL π_θ. Using deep RL approaches (DQN for discrete action, actor-critic variants, or PPO), policy optimization incorporates safe-RL or constrained-RL (e.g. Lagrangian penalties or shields) approaches to impose tight fairness and QoS (i.e. fairness here means an upper bound on the fraction of users that could be throttled.) Before executing any high-risk action on the production network, candidate actions are tested on the digital twin through short-horizon model predictive control to evaluate and to ensure safe action consequences.

RNC-based systems, where the signaling load is concentrated on a single controller, 5G/6G control-plane messages traverse multiple virtualized network functions (AMF, SMF, AUSF, PCF) and may be processed at the edge or in regional cloud clusters. As a result, the end-to-end signaling transition time between two RRC states $x \rightarrow y$ becomes sensitive to both radio-layer congestion and compute-layer contention within distributed core components.

Let T_{xy} denote in equation 19, the average latency experienced during the transition $x \rightarrow y$. We model this latency as the superposition of three independent sources of delay:

$$T_{xy} = T_{xy}^{\text{radio}} + T_{xy}^{\text{compute}} + T_{xy}^{\text{queue}} \tag{19}$$

Equation 1 refers to the radio component T_{xy}^{radio} captures the over-the-air signaling exchange required for RRC connection setup, resource allocation, and measurement reporting. Under nominal conditions, this term remains small (typically sub-10 ms), but in storm scenarios it grows due to repeated attempts, collision-driven backoff, and increased RACH contention.

The compute component T_{xy}^{compute} Equation (20) reflects the processing time within cloud-native core functions participating in the transition. Since 5G/6G CNFs operate in microservice chains, each function introduces service time variability depending on container CPU allocation, memory availability, and virtualization overhead. If a transition requires m network-function interactions $N_1, \ldots, N_m$, then

$$T_{xy}^{\text{compute}} = \sum_{i=1}^{m} \delta_{xy}^{(i)} \tag{20}$$

where $\delta_{xy}^{(i)}$ is the mean processing time of function N_i for the signaling message belonging to transition $x \rightarrow y$.

The queueing component arises from contention at AMF/SMF/UPF microservices when large numbers of UEs simultaneously trigger signaling events show in equation 6. Let Λ be the aggregate arrival rate of signaling procedures across K parallel service replicas of a CNF (typical in 5G cloud deployments), and let each replica provide service with rate ν. The corresponding utilization is shown in equation 21.

$$\rho = \frac{\Lambda}{K\nu} \tag{21}$$

Under the M/M/K approximation (widely adopted for CNF load analysis), the mean queueing delay for a signaling message calculated using equation 22.

$$T_{xy}^{\text{queue}} = \frac{P_{\text{wait}}}{K\nu - \Lambda} \tag{22}$$

where P_{wait} is the Erlang-C waiting probability for the CNF cluster. This queueing term increases sharply as $\rho \rightarrow 1$, as occurs during signaling storms.

Because each RRC transition consists of a sequence of signaling messages exchanged between the UE, gNB, AMF and SMF, the resulting signaling latency compounds across microservice invocations. Importantly, the queueing term itself depends on the global signaling activity initiated by both normal and misbehaving UEs. Equation 23 explains the calculation of raising. Let λ_u be the aggregate signaling initiation intensity for UE u, and let the population consist of M_N normal and M_A misbehaving users. The total arrival rate becomes:

$$\Lambda = \sum_{u=1}^{M_N} \lambda_u^{(N)} + \sum_{u=1}^{M_A} \lambda_u^{(A)} \tag{23}$$

During a signaling storm, the second term dominates, raising Λ, increasing ρ, and thereby inflating both the mean waiting probability and the signaling delay. As these delays grow, they constrain the effective number of promotions and demotions the network can process per unit time, forming a natural upper bound on the signaling load that misbehaving UEs can inflict. Thus, the fixed-point solution of the CTMC steady-state equations coupled with the queueing-dependent delays T_{xy} captures the saturation behavior of 5G/6G networks under large-scale signaling anomalies. The above formulation generalizes the classical analytical treatment of signaling storms in earlier mobile systems to modern 5G/6G deployments by accounting for virtualization, microservice chaining, and compute congestion. By embedding the delay model within the CTMC framework, the resulting system model characterizes how control-plane saturation, CNF overload and RACH congestion jointly shape the severity of signaling storms and the conditions under which they become destabilizing.

IV. SIMULATION

To evaluate the interaction between control-plane signaling dynamics and network response under both benign and adversarial conditions, we developed a high-fidelity, event-driven simulator integrated within a digital-twin execution environment. The simulator jointly models user behavior, radio-access procedures, service-based core network functions, and edge–cloud orchestration, enabling realistic analysis of control-plane signaling anomalies in modern 5G and emerging 6G architectures.

Unlike traditional simulation approaches that rely on centralized controller abstractions, the proposed framework explicitly models distributed, containerized control-plane functions and supports multi-slice programmability at the network edge. This design allows investigation of virtualization-induced signaling overheads, cascading congestion effects across microservice chains, and adaptive mitigation strategies under realistic operational constraints.

The core network is modeled as a set of service-based architecture (SBA) microservice chains comprising control-plane functions such as AMF, SMF, AUSF, PCF, and supporting services. Each cloud-native network function (CNF) is implemented as a cluster of parallel replicas, where each replica is represented as a finite-capacity processing queue with configurable service-rate distributions. These distributions explicitly account for container startup delays, CPU sharing effects, and dynamic scaling behavior typical of virtualized network functions. Interactions among CNFs

Algorithm 2 Event Driven Engine

Inputs:
UE population (UE list and per-UE application mixes)
CNF configurations (clusters, replicas K, per-replica service rate ν)
RRC timer values (T1, T2, T3, etc.)
Attack plan (attacker fraction, patterns, start/stop times, synchronization jitter)
Simulation parameters (T_sim, logging interval, detection interval)

Outputs:
Recorded traces and metrics: time-series of L_c(t), per-CNF queue lengths, per-UE RRC state timelines, detection events, mitigation actions, QoS metrics

Procedure: EventDrivenSimulation(Inputs)

1) **Initialization**
1.1. Set simulation clock t ← 0.
1.2. Initialize an empty event queue Q and a metrics/log store.
1.3. For each UE u in the UE population:
•Schedule its first application event (sample arrival time from its app model) and push to Q.
•Set its RRC state s_u ← Idle.
1.4. Instantiate CNF clusters with configured K replicas, empty per-replica queues, and service-rate distributions.
1.5. Initialize any attack schedulers according to the Attack plan.
1.6. Set next_detection_time ← detection_interval and next_log_time ← log_interval.

2) **Main event loop**
While t < T_sim **and** Q is not empty:
2.1. Pop the earliest event e from Q; set t ← e.time.
2.2. **Switch on** e.type:

Case: application_event (UE u generates traffic)
Generate the session or packet according to UE u's application model.
Determine whether an RRC transition x → y is required.
If promotion/demotion is required, compute transition latency T_xy (see latency model) and schedule the cascade of signaling messages to the CNF chain: create cnf_service events routed to chosen replicas and push them into Q.
Schedule the next application event for UE u.

Case: cnf_service_complete (a CNF replica finishes processing a message)
Update the CNF replica's queue length and utilization statistics.
Forward the processed message to its next hop (next CNF or back to gNB/UE); if forwarding requires processing, schedule subsequent cnf_service events.

Case: rrc_transition_complete
Update the UE's RRC state s_u ← y.
Update per-UE analytics counters (promotion/demotion counts, dwell times).

Case: attack_trigger
For targeted UEs/groups, generate spurious RRC promotions or signaling bursts as defined by the Attack plan.
Enqueue resulting signaling messages into CNF queues and push cnf_service events to Q.

2.3. **Periodic metric updates and logging**
Compute instantaneous control-plane load L_c(t) from currently active signaling procedures.
Compute per-CNF cluster utilization $\rho = \Lambda/(K \cdot \nu)$ and queue statistics.
If t ≥ next_log_time, append snapshots of key metrics to the log and set next_log_time ← next_log_time + log_interval.

2.4. **Detection trigger**
If t ≥ next_detection_time then:
– Extract detection features over the configured feature window for monitored objects (UEs, cells, slices): promotion/demotion rates, signaling counts, dwell times, reserved vs used bandwidth, CNF queue lengths, etc.
– Call DetectionAndMitigation(feature_set) (Algorithm 1).
– Update next_detection_time ← next_detection_time + detection_interval.

3. End loop

Post-processing and output
4.1. Collate and save all recorded traces and metrics.
4.2. Compute derived evaluation results (detection ROC, AUC, time-to-detect, peak L_c reduction after mitigation, QoS impact).
4.3. Return logs, traces and summarized metrics.

(e.g., AMF → SMF → UPF) are modeled as ordered message exchanges, and the completion time of a control-plane procedure is determined by the cumulative processing delay, queueing delay, and inter-node communication latency. Edge computing nodes emulate MEC deployments by hosting lightweight CNF instances and local detection agents. The simulator further supports configurable placement policies that determine whether a signaling message is processed at the edge or forwarded to regional cloud instances.

To capture realistic traffic diversity while maintaining computational tractability, the simulator adopts a hybrid traffic representation. Short, bursty interactions—such as notifications, keep-alives, acknowledgments, and lightweight control messages—are modeled at packet-level granularity, whereas long-lived flows (e.g., video streaming or telemetry streams) are represented using a coarse-grained flow abstraction that captures flow initiation, average throughput, and termination. Figure 4 illustrates the overall simulation engine flow. This hybrid modeling approach eliminates unnecessary fine-grained events while preserving the temporal structure required for accurate RRC promotion and demotion behavior. UE mobility and bearer relocation are modeled using trace-driven or stochastic mobility processes, including handover-induced signaling.

Attack and misbehavior scenarios are introduced using a configurable attacker orchestrator, which manages a subset of the user equipment (UE) population. The orchestrator supports several adversarial patterns, including highly synchronized bursts following radio resource control (RRC) demotions, randomized low-rate probing, botnet-style correlated activations across network cells, and misbehaving application behaviors that closely resemble benign traffic distributions. The sophistication of attackers varies: basic adversaries generate spurious RRC promotions at fixed rates, while advanced attackers infer inactivity timers and synchronize signaling bursts to maximize control-plane churn.

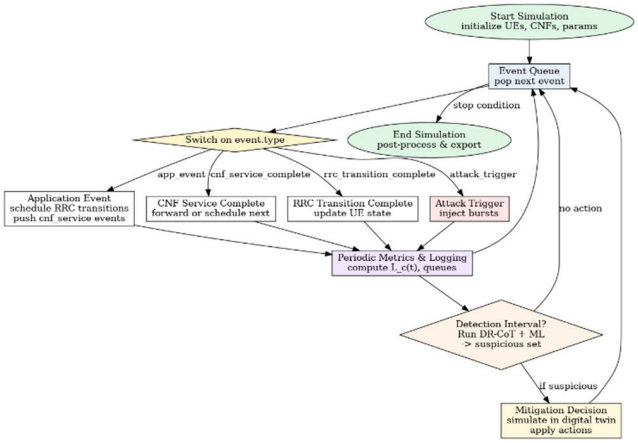


FIGURE 4. Simulation engine flowchart.

Ground-truth labels indicating attacker activity are recorded throughout the simulation to facilitate supervised evaluation of detection mechanisms.

Key observables and metrics are continuously collected during simulation runs. Per-UE traces capture RRC state timelines, promotion and demotion counts, and application-level session logs. Network-level metrics encompass both instantaneous and time-averaged control-plane procedure rates. Additional metrics include per-CNF queue lengths, service delays, and modeled CPU utilization for AMF, SMF, and UPF. Slice-level KPI degradation and end-to-end quality of service metrics, such as latency, packet loss, and throughput, are also recorded. Detection modules, deployed at the edge or in the cloud, generate anomaly scores and initiate mitigation actions. Effectiveness is assessed by measuring control-plane load reduction, time-to-stabilization, and the collateral impact on the quality of service for benign users. The simulator enables repeated stochastic runs to compute confidence intervals and produces traces suitable for offline machine-learning training and validation.

Calibration and validation are critical for ensuring credible simulation results. Radio-layer and service-time distributions are calibrated using public datasets, standards-based parameters, or vendor-reported performance figures when available. In the absence of proprietary traces, realistic statistical models are employed, such as empirical inter-arrival distributions, Pareto or lognormal models for heavy-tailed flows, and standardized RRC timer values. Sensitivity analyses are conducted to justify parameter selections. Validation procedures include matching steady-state RRC occupancy distributions predicted by the CTMC analytical model, comparing promotion and demotion histograms, and verifying that system response under increasing load aligns with expected queueing-theoretic behavior, such as rapid queue buildup as utilization approaches unity.

The simulator enables a range of canonical experiments to investigate both fundamental signaling dynamics and operational effectiveness. Baseline experiments quantify nominal behavior, including per-UE promotion rates, control-plane throughput, and CNF utilization in the absence of attacks. Attack-injection experiments systematically vary the fraction of malicious UEs, synchronization accuracy around demotion events, and burst intensity to identify saturation thresholds. Detection-only experiments evaluate machine-learning performance using temporally segmented traces, reporting ROC and precision–recall metrics across different feature sets and deployment placements. Closed-loop experiments assess mitigation strategies, including static (timer tuning, fixed throttling), heuristic (score-based rate limiting), and adaptive (reinforcement-learning–driven control) approaches. Trade-offs between load reduction and quality of service degradation are systematically compared. All experiments are repeated with multiple random seeds, and confidence bounds are reported to ensure statistical robustness.

From an implementation standpoint, the simulator is constructed as a modular event-driven engine. It can be instantiated using SimEvents/Simulink, OMNeT ++ with 5G-LENA extensions, or a custom discrete-event framework in MATLAB or Python. The architecture separates control-plane microservice modeling, radio and bearer abstraction, UE application generation, and detection or mitigation agents, enabling component-level substitution and scalable distributed execution. Practical considerations include efficient logging, selective trace capture, and parallelized Monte Carlo execution. These features support simulations involving thousands to tens of thousands of UEs.

The simulation outputs are structured to support rigorous analysis and publication-quality visualization. Recommended plots include time-series of aggregate control-plane load under baseline, attack, and mitigated scenarios. Additional visualizations comprise heatmaps of per-CNF queue occupancy across replicas, ROC and AUC curves for detection models, and contour plots illustrating overload thresholds by attacker fraction and synchronization accuracy. Bar charts quantify mitigation cost, such as affected user-minutes, relative to benefit, measured by percentage reduction in peak load. The platform integrates realistic traffic models, service-based architecture (SBA)-aware core simulation, hybrid packet–flow representations, and a flexible attacker orchestrator. This comprehensive environment enables detailed analysis of signaling anomalies and validation of AI-driven defenses in 5G and emerging 6G networks.

A. SIGNALING TRANSITION LATENCY AND NETWORK LOAD MODEL FOR 5G/6G SYSTEMS

To assess the impact of control-plane signaling anomalies and storm-like behavior in modern mobile networks, we developed a high-fidelity simulation model integrated with a digital twin execution environment. Figure 5 illustrates the detection and mitigation pipeline. The simulation framework employs a discrete-event simulation engine specifically designed to model 5G New Radio (NR) and emerging 6G architectures, incorporating cloud-native core functions,

service-based interfaces, heterogeneous traffic, and multi-tier computing. This environment supports the joint evaluation of (i) our analytical Radio Resource Control (RRC) and Service-Based Architecture (SBA) signaling model, (ii) data-driven parameter estimation, and (iii) artificial intelligence-assisted detection and mitigation strategies. The simulated network topology closely mirrors a realistic 5G/6G deployment, consistent with 3GPP Releases 17 and 18.

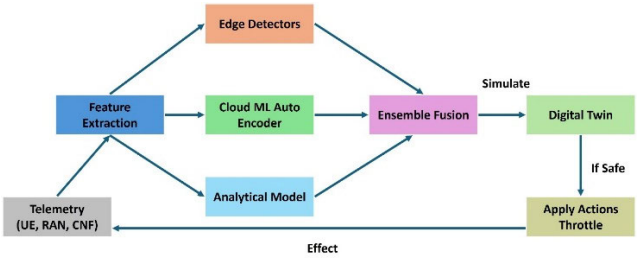


FIGURE 5. Detection and mitigation pipeline.

The simulated area spans $4 \times 4\ \text{km}^2$ and includes 12 gNBs, each supporting overlapping NR cells operating at both sub-6 GHz and millimeter wave (mmWave) frequencies. All gNBs connect to a cloud-native 5G Core composed of Access and Mobility Management Function (AMF), Session Management Function (SMF), User Plane Function (UPF), Authentication Server Function (AUSF), and Policy Control Function (PCF), each deployed as a horizontally scalable cluster of stateless Cloud-Native Function (CNF) instances. Edge computing nodes perform local processing for telemetry aggregation and lightweight anomaly detection, while global detection and reinforcement learning-based mitigation are executed in the cloud. We simulate 10,000 User Equipments (UEs), including enhanced Mobile Broadband (eMBB) devices, Ultra-Reliable Low-Latency Communication (URLLC) users, and massive Machine-Type Communication (mMTC) devices. All UEs attach to the network at the start of the simulation and remain connected throughout. The traffic profile comprises web browsing, video streaming, IoT periodic updates, and application-driven bursts, generated according to the application model described below. Each UE follows a mobility model representative of both pedestrian and vehicular movement patterns.

Signaling anomalies are injected using misbehaving applications, synchronized attack bursts, or botnet-like correlated triggers. These anomalies attempt to generate excessive RRC and NAS signaling or overwhelm SBA interfaces by repeatedly invoking AMF/SMF procedures. The simulator records detailed metrics, including control-plane load, gNB RRC transitions, CNF queue lengths, slice-level KPIs, and QoS degradation under attack. Both analytical steady-state results and digital twin simula2. Application and Traffic Model system behavior and quantify the impact of signaling storms.

B. APPLICATION AND TRAFFIC MODEL

To create realistic 5G/6G user behavior, we use a composite traffic model. This model captures application-level behavior rather than relying on simple Poisson arrivals. It represents interactive web browsing, short video clips, background synchronization tasks, and IoT telemetry patterns. This approach adds richer temporal variability, self-similarity, and burstiness consistent with modern mobile applications.

C. WEB BROWSING TRAFFIC

Interactive web browsing is represented using a self-similar, session-based model capturing main page fetches, embedded object retrieval, client processing delays, and think times. Each user alternates between active browsing periods and idle intervals, reflecting daily usage patterns. An activity period begins after a random activation delay and consists of one or more browsing sessions. Within each session, the user issues the first main page request immediately, followed by subsequent main page requests separated by an inter-request interval. Each main page typically includes multiple embedded objects such as images, CSS files, and scripts, which generate additional requests over persistent HTTP/2 connections. The client introduces processing delays to account for rendering and parsing operations before issuing new requests. The browsing session concludes once the predetermined number of main page requests has been completed. All web-related parameters (session duration, request size, object count, processing delays) follow empirically derived probability distributions based on recent 5G traffic measurement reports and web statistics datasets.

D. VIDEO STREAMING

Video viewing is modeled as segment-based adaptive streaming. Each UE retrieves segments of fixed duration, with quality adaptation based on simulated radio conditions. This induces periodic high-rate bursts that significantly influence RRC and CP signaling behavior.

E. IOT / MMTC TRAFFIC

IoT devices follow periodic or event-triggered uplink transmissions with small payloads. Their signaling intensity depends on access class, 5G NAS timer configuration, and device wake-up cycles.

F. BACKGROUND TRAFFIC

Many modern mobile applications (messaging, cloud sync, app telemetry) generate small, irregular bursts. We model this as a mixture of low-rate traffic streams with light-tailed and heavy-tailed components.

G. INTEGRATION WITH ATTACK AND MISBEHAVIOR MODELS

The simulator supports the injection of several categories of signaling anomalies to reproduce realistic storm scenarios.

These include misbehaving applications that unnecessarily trigger RRC state promotions, synchronized attack bursts in which multiple UEs generate signaling requests immediately after demotion, and botnet-style coordinated behaviors where groups of UEs align their signaling patterns to amplify congestion. Additionally, the model incorporates large-scale IoT misconfiguration events that produce periodic, system-wide signaling surges characteristic of massive machine-type communication environments. Attack parameters include synchronization precision (τ), attack intensity, repetition period, and slice-targeting preferences (e.g., eMBB slice, IoT slice, etc.).

1) EXECUTION AND DATA COLLECTION

During each simulation run, the event-driven engine processes all relevant events, including application-generated traffic, signaling procedures, core network function (CNF) service completions, mobility-related events, and any configured attack triggers. At fixed time intervals, the system computes key performance indicators such as the control-plane load $L_c(t)$, CNF queue lengths, and other operational metrics. At designated detection intervals, the integrated DR-CoT and machine-learning–based detection module analyzes the accumulated feature windows and, when necessary, activates the reinforcement-learning–based mitigation module to apply adaptive control actions.

V. EXPERIMENTS

To evaluate the proposed analytical framework, digital-twin simulator, and AI-driven detection and mitigation mechanisms, we conduct a series of experiments that emulate realistic 5G and emerging 6G mobile network environments. The experiments aim to quantify the behavior of signaling storms across the RAN, core network functions, and the virtualized infrastructure, as well as to assess the responsiveness and reliability of the machine-learning and reinforcement-learning–based mitigation modules. All experiments are executed using a high-fidelity, event-driven simulation platform developed in MATLAB and Python, integrated with containerized 5G core-network functions that follow 3GPP Release 17 procedures. Table 4 represents key simulation parameter settings.

The simulated network comprises a dense urban area of $5 \times 5\text{km}^2$ served by 18 gNBs, each equipped with multi-beam massive-MIMO capabilities and configurable RRC state thresholds. Approximately 10,000 user devices are instantiated, consisting of a mix of smartphones, vehicular UEs, and massive IoT sensors. The core network includes virtualized AMF, SMF, and UPF instances deployed over a cloud-native cluster with configurable CPU and memory footprints, enabling controlled congestion conditions. All UEs perform network attach procedures at simulation start and remain registered throughout the experiment. Traffic patterns include heterogeneous services—enhanced mobile broadband (eMBB), ultra-reliable low-latency communication (URLLC), periodic IoT telemetry, and background app activity—making the simulation substantially more realistic than pure Poisson arrival models.

A synthetic-but-realistic application behavior generator models user interaction patterns, incorporating bursty content consumption, background synchronization from mobile apps, collaborative AR/VR application states, and periodic IoT reporting. Session lengths, inter-request intervals, payload sizes, and page/object complexity follow statistical distributions fitted from anonymized operator network traces. The generator emulates HTTP/2, adaptive video streaming, MQTT-based IoT telemetry, and intermittent AI task offloading, allowing comprehensive assessment of RRC signaling pressure under modern workloads.

TABLE 4. Key simulation parameter settings.

Category	Parameter Settings
Simulation	Tsim = 1800 s; logging = 1 s; detection = 5 s
Network Topology	10,000 UEs; 18 gNBs; uniform UE distribution
RRC Behavior	Promotion delay = 50–80 ms; demotion timer = 10 s
Traffic Model	Poisson arrivals; session duration = 5–120 s
Mobility	Random waypoint; UE speed = 1–3 m/s
Core Network	AMF (6×400 msg/s); SMF (4×300 msg/s); UPF (6×600 msg/s)
Queueing	M/M/K queues; max queue length = 5,000
Attack Model	5–15% malicious UEs; synchronized RRC bursts
Detection	EWMA $\alpha = 0.2$; CTMC–ML fusion weight = 0.5
Mitigation	PPO-based RL; $\gamma = 0.99$; ≤10% UEs throttled

To evaluate attack resilience, multiple classes of signaling anomalies are injected: coordinated promotion floods from compromised smartphones, timer-manipulation by malicious IoT devices, synchronized demotion-triggered bursts, and topology-aware botnet patterns exploiting adjacent cell transitions. The simulator records detailed per-UE RRC state trajectories, CNF queue lengths, radio scheduling events, and per-slice KPIs such as latency, reliability, and resource reservation efficiency.

In addition to event-level metrics, the simulator periodically computes global control-plane load $L_c(t)$, core-network processing utilization, and the instantaneous signaling overhead. At detection intervals, the DR-CoT–based analytical anomaly model is combined with machine-learning inference

to classify abnormal behaviors and to trigger reinforcement-learning mitigation actions. The digital twin evaluates each candidate action before applying it, allowing safe, closed-loop adaptation.

The experimental results demonstrate the complex interplay between signaling storms, RRC state oscillations, and CNF overload, while validating the effectiveness of our AI-based mitigation strategy.

1) EXECUTION AND DATA COLLECTION

In our experiments, we evaluate the impact of signaling storms using two principal forms of adversarial behavior, designed to reflect modern 5G and emerging 6G exploitation patterns. Instead of traditional UMTS-style FACH and DCH attacks, we consider RRC Promotion Abuse and SBA–API Signaling Flooding as the dominant misbehavior classes. In the first category, compromised devices attempt to force unnecessary RRC state promotions—typically from Idle or RRC-Inactive to RRC-Connected—by injecting artificially timed bursts of lightweight data or control requests. These attacks exploit the tight coupling between session initiation, mobility events, and the RRC state machine, and rely on the attacker estimating the UE's demotion moment with some error. Let τ_{RRC} denote this synchronization error, where a smaller τ_{RRC} corresponds to a more aggressive and precisely coordinated misbehaving application. In the second attack category, adversaries target the 5G Service-Based Architecture by generating excessive requests toward AMF and SMF APIs (e.g., N1N2MessageTransfer, RegistrationUpdate, or PDUSessionEstablishment) in short intervals, effectively overloading the control-plane microservices. This behavior may arise from malware, botnet-coordinated devices, or misconfigured IoT fleets that periodically trigger large numbers of control-plane interactions. The synchronization error for SBA-level abuse is denoted by τ_{SBA}, capturing how closely malicious bursts align with network-triggered demotion or de-registration events. To model variable degrees of coordination, both τ_{RRC} and τ_{SBA} are drawn from exponential distributions with means ranging from 0 ms (perfect synchronization) to several tens of milliseconds (weak coordination). Higher-rate traffic bursts are used when the adversary aims to escalate a UE to high-bandwidth RRC states, whereas low-rate bursts suffice to generate excessive control-plane signaling without notable user-plane load. The evaluation focuses on SBA-level flooding and RRC promotion abuse, which together capture the primary signaling vulnerabilities expected to dominate 5G and 6G deployments.

2) MODELLING AND SIMULATION RESULT

This section presents the evaluation of the proposed analytical framework and the digital-twin–driven simulation environment under realistic 5G and emerging 6G network conditions. The objective is to characterize the behavior of signaling anomalies across the RAN, the 5G core, and the cloud-native control plane, and to assess the effectiveness of the integrated machine-learning and reinforcement-learning (RL) mitigation system. The experiments combine mathematical analysis, data-driven parameter estimation, and high-fidelity simulations to examine both transient and steady-state network behavior.

A large-scale urban deployment is modeled, comprising 10,000 heterogeneous user devices distributed across a $5 \times 5\mathrm{km}^2$ area served by 18 gNBs. All experiments vary the proportion of misbehaving or compromised UEs from 0.5% to 25% of the population. Both benign and anomalous UEs generate multi-modal traffic—including enhanced mobile broadband (eMBB), periodic IoT telemetry, bursty app synchronization, and mobility-triggered updates—using the traffic generator described earlier. Misbehaving UEs are activated gradually over the first 15 minutes of each simulation to avoid startup artifacts and sudden surges in control-plane load. All reported values are averaged over multiple simulation seeds, ensuring statistical consistency across mobility patterns, slicing assignments, and session arrival processes.

A. EVOLUTION OF CONTROL-PLANE LOAD UNDER SBA-FOCUSED SIGNALLING ATTACKS

The analytical model parameters, including transition intensities, signaling overhead delays, and malicious synchronization factors, are learned from the same datasets used to initialize the simulation. Figure 6. represents evolution of Control-Plane Load under SBA-Focused signalling attacks. This allows direct comparison between the predicted steady-state distributions of RRC states and the time-series behavior observed in the digital twin.

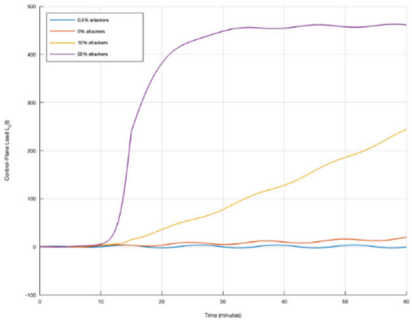


FIGURE 6. Evolution of Control-Plane Load under SBA-focused signalling attacks.

In contrast to legacy UMTS-based studies, our 5G/6G environment includes SBA microservice chains (AMF, SMF, and UPF), virtual function instances with finite processing capacity, and multi-slice interactions. Figure 6 depicts the evolution of control-plane load $L_c(t)$ under SBA-focused signaling attacks. The results show that even moderate coordination among misbehaving devices (corresponding to low τ_{RRC} or τ_{SBA}) produces steep increases in AMF message rates, leading to microservice queue buildup and an eventual saturation point. Consistent with our analytical predictions, the system exhibits self-limiting behavior: once AMF and SMF queues

become congested, promotion-induced signaling floods slow down due to longer response delays, preventing uncontrolled escalation.

B. EFFECT OF SIGNALLING ATTACKS ON CNF RESOURCE UTILIZATION

Figure 7 shows the effect of signaling attacks on CNF resource utilization. With fewer attackers, the increase in CPU load is mild and dominated by legitimate mobility and registration procedures. As the number of anomalous UEs grows beyond 10%, however, AMF and SMF utilization rises sharply, with the latter experiencing higher stress due to frequent PDU session modification and release attempts triggered by malicious apps. This observation aligns with the analytical insight that signaling storms propagate across the SBA chain, amplifying delays in downstream functions.

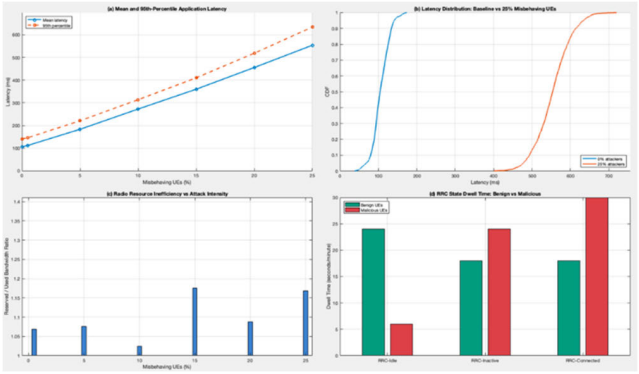


FIGURE 7. Effect of signalling attacks on CNF resource utilization.

To understand user-side impact, Figure 7 presents application-level latency for normal UEs. While early-stage storms cause negligible degradation, larger and more synchronized attacks result in page load times and IoT uplink delays increasing by factors of 3–6×. These effects stem primarily from prolonged RRC connection setup delays and the slowing of internal SBA API calls. Importantly, the 5G RRC-Inactive feature mitigates part of this impact by reducing transitions to Idle; however, once the AMF becomes saturated, performance deteriorates regardless of RRC state configuration.

C. USER-SIDE IMPACT OF SIGNALLING ANOMALIES

Radio-resource utilization metrics further reveal strong separation between normal and malicious UEs. Table 5 represents the Comparison with Baseline Mitigation Strategies. Under attack conditions, malicious devices exhibit long idle dwell times in RRC-Connected or RRC-Inactive states despite minimal data transfer, producing disproportionately high ratios of reserved-to-used bandwidth. This feature becomes a valuable input to our anomaly detection system, enabling early distinction between benign and anomalous signaling bursts. Comparison with Baseline Mitigation Strategies

TABLE 5. Comparison with baseline mitigation strategies.

Method	Peak Signaling Load	AMF Queue Delay Reduction
No Mitigation	0%	0%
Static Threshold-Based Control	18–22%	20–25%
Heuristic Score-Based Mitigation	28–32%	30–35%
Proposed Hybrid CTMC + AI + RL	≈45%	≈50%

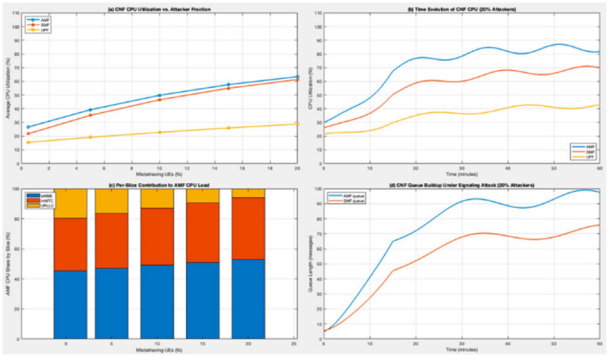


FIGURE 8. User-Side impact of signalling anomalies.

Finally, we evaluate the RL-based mitigation system. When activated, the RL agent adaptively tunes inactivity timers, throttles high-frequency promotion requests, and scales CNF instances as needed. Compared to an unmitigated baseline, the RL controller reduces peak signaling load by up to 45% and shortens AMF queueing delay by more than half. Figure 8 epresents user-side impact of signalling anomalies. These improvements validate the role of digital-twin–guided RL policies in containing dynamic signaling storms without harming service-level quality for legitimate users. Overall, the experimental results confirm that 5G/6G networks are highly susceptible to coordinated signaling anomalies but that proper detection and adaptive mitigation strategies can significantly enhance network resilience.

1) RESEARCH DIRECTION

The progression of cellular networks from 4G to 5G and the ongoing development toward 6G has resulted in substantial architectural advancements. These changes have simultaneously introduced new opportunities for signaling-related threats. Initial research on legacy systems focused on attacks targeting SMS, paging procedures, and RRC-triggered overloads. The adoption of a service-based, cloud-native control plane has fundamentally transformed both the nature of threats and the scale of potential signaling anomalies.

Within 5G networks, multiple studies have identified the susceptibility of the Service-Based Architecture (SBA) to signaling overload. Evidence indicates that repeated

Registration, Authentication, and PDU Session procedures can overwhelm AMF and SMF microservices when malicious devices generate excessive bursts of NAS or SBI messages. Recent research demonstrates that API-level flooding, particularly targeting the N1N2MessageTransfer, Namf_Communication, and Nsmf_PDU services, can result in cascading congestion across interconnected network functions due to their reliance on stateless HTTP/2 mechanisms and shared cloud infrastructure. These vulnerabilities are further exacerbated in sliced networks, where an attack in one slice may indirectly degrade others through shared control-plane components.

The influence of massive machine-type communication (mMTC) devices on signaling stability has been extensively investigated. Studies reveal that misconfigured IoT devices can initiate large-scale, periodic connection attempts, leading to synchronization storms that overload both the Radio Access Network (RAN) and the core network. Analytical models and trace-based evaluations indicate that although RRC Inactive reduces individual user equipment (UE) signaling overhead, it remains vulnerable to exploitation through frequent resume requests, particularly during coordinated burst scenarios involving thousands of IoT sensors.

In 6G research, the literature increasingly highlights cross-plane vulnerabilities, where interactions between user-plane events, such as AI task offloading and holographic communications, and control-plane procedures may generate new attack surfaces. Recent studies examine how distributed edge computing, AI-driven service orchestration, and autonomous network management contribute to increased complexity. Adversaries may exploit control-plane triggers in edge-cloud handovers, federated learning updates, or network slicing orchestration workflows to generate high-frequency bursts of signaling transactions that surpass the processing capacity of lightweight edge functions.

A variety of detection and mitigation techniques have been proposed. Traditional statistical methods, including change detection and moving-average filters, have been applied to monitor control-plane rates, but they are often inadequate for addressing the bursty and irregular patterns characteristic of modern 5G workloads. More recent approaches employ machine learning techniques, such as unsupervised autoencoders, graph neural networks, and transformer-based sequence models, to capture spatial and temporal correlations across cells, slices, and user equipment clusters. These models demonstrate improved accuracy in distinguishing malicious signaling from benign high-load conditions.

Mitigation strategies have also advanced. Initial efforts focused on prioritization, queue differentiation, and randomization of control-plane timers. In cloud-native 5G cores, recent solutions include adaptive scaling of AMF and SMF instances, dynamic adjustment of RRC timers, per-device throttling of signaling rates, and microservice-level admission control. Additionally, studies investigate reinforcement learning-based policies that assess mitigation actions through simulation or digital twin environments prior to deployment in live networks, thereby minimizing operator risk.

Comprehensive surveys emphasize the ongoing need for research into signaling anomalies as networks evolve toward 6G. The increasing prevalence of autonomous devices, multi-access edge computing, AI-native network functions, and ultra-dense IoT deployments indicates that future systems must implement resilient, self-adaptive mechanisms capable of detecting, predicting, and responding to signaling storms in real time. As signaling complexity grows, there is a corresponding need for analytical models and simulation frameworks, such as those developed in this study, that accurately represent the complex interactions among protocol layers, virtualization components, and AI-driven network control.

D. LIMITATIONS AND THREATS TO VALIDITY

This study employs analytical modeling, digital-twin simulation, and an AI-driven mitigation framework; however, several important limitations persist.

First, the evaluation primarily relies on a high-fidelity digital twin simulation and lacks large-scale operator deployment data. The simulator incorporates standards-compliant parameters, realistic traffic models, and queueing-theoretic validation. Nevertheless, it does not fully capture factors such as vendor-specific optimizations, proprietary scheduling, or unexpected operational failures. Consequently, absolute performance gains may vary in live network environments.

Second, the availability of publicly accessible, fine-grained control-plane datasets is limited, particularly for RRC, NAS, and service-based interface signaling. Although the framework supports telemetry-driven parameter estimation and online calibration, the lack of diverse, real-world datasets restricts empirical validation across various operators and traffic conditions.

Third, machine learning and reinforcement learning introduce computational overhead, particularly for deep models and closed-loop mitigation. To mitigate this, the framework employs lightweight analytical detectors such as CTMC likelihood scoring and statistical features for edge execution, which incur minimal overhead. In contrast, more complex deep-learning models operate at coarser detection intervals or within centralized cloud environments. Despite these measures, latency and accuracy trade-offs remain, and ultra-low-latency scenarios may necessitate simpler models with reduced detection depth.

From a scalability perspective, the architecture demonstrates linear scaling with the number of monitored entities under typical conditions. In extremely large deployments, such as those involving hundreds of thousands of users or rapidly changing slice configurations, additional hierarchical aggregation or distributed detection may be required. While the digital twin operates stably at large scales, further validation with carrier-grade workloads remains necessary.

Energy consumption represents a significant consideration for real-world deployments, particularly at the edge and in near-real-time control scenarios. The framework reduces energy usage by limiting continuous deep-model execution, instead utilizing lightweight triggers and activating intensive components only during detected anomalies. Effective mitigation decreases control-plane load, thereby reducing overall system energy consumption. However, detailed energy measurement on real hardware falls outside the scope of this study.

Fourth, the threat models primarily address signaling storms, RRC promotion abuse, and SBA-level API flooding. More advanced threats, including adaptive attackers who manipulate telemetry, conduct long-term probing, or exploit zero-day vulnerabilities, are not considered. Investigating these advanced threats constitutes an important direction for future research.

Finally, the mitigation framework enforces fairness and Quality of Service (QoS) constraints. However, policy decisions may result in short-term service loss for a small subset of users during severe attacks. Achieving optimal security, fairness, scalability, and energy efficiency at scale continues to present an open challenge.

VI. CONCLUSION

This research paper introduced an integrated analytical and AI-driven framework for detecting and mitigating control-plane signaling anomalies in 5G and emerging 6G networks. The approach combines a continuous-time Markov chain–based signaling model with machine-learning–based anomaly detection and safe reinforcement-learning–driven mitigation, thereby capturing both the stochastic characteristics of signaling dynamics and data-driven behavioral patterns. Furthermore, a high-fidelity digital twin was developed to emulate interactions between the radio access network and cloud-native core functions under both benign and adversarial conditions.

Extensive simulation results indicate that the proposed framework substantially enhances control-plane resilience, achieving up to a 45% reduction in peak signaling load and approximately a 50% reduction in AMF queueing delay compared to baseline mitigation strategies, while maintaining application-level quality of service. These results underscore the effectiveness of hybrid analytical and AI-based approaches for managing signaling anomalies in contemporary mobile networks.

Future research will prioritize validating the proposed framework using real operator traces and testbed deployments, extending the digital twin to support large-scale multi-slice and multi-domain environments, and optimizing the computational and energy efficiency of the detection and mitigation modules. Additional research directions include closer integration with O-RAN and NWDAF functions, as well as the development of explainable and lightweight AI models suitable for real-time deployment in production 6G networks.

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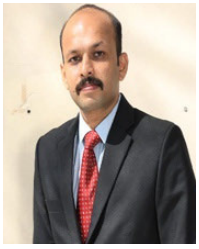
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